

Implementation

Step 1: Create and Select Database

Command:

```
mysql> CREATE DATABASE EmpDB;  
mysql> USE EmpDB;
```

Output:

```
Query OK, 1 row affected (0.01 sec)  
Database changed
```

Step 2: Create Tables

2a. Create Employees Table

Command:

```
mysql> CREATE TABLE Employees (  
->     EmpID      INT AUTO_INCREMENT PRIMARY KEY,  
->     Fname      VARCHAR(20),  
->     Lname      VARCHAR(20),  
->     Dname      VARCHAR(20),  
->     DOJ        DATE,  
->     Salary     DECIMAL(10,2)  
-> );
```

Output:

```
Query OK, 0 rows affected (0.02 sec)
```

2b. Create Employee_Audit Table

Command:

```
mysql> CREATE TABLE Employee_Audit (  
->     AuditID     INT AUTO_INCREMENT PRIMARY KEY,  
->     EmpID       INT,  
->     Fname       VARCHAR(20),  
->     Lname       VARCHAR(20),  
->     Dname       VARCHAR(20),  
->     DOJ         DATE,  
->     Salary      DECIMAL(10,2),  
->     ActionType  VARCHAR(10),  
->     ActionDate  TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
-> );
```

Output:

```
Query OK, 0 rows affected (0.02 sec)
```

Step 3: Create Triggers for Audit Logging

3a. AFTER INSERT Trigger

Command:

```
mysql> DELIMITER $$
mysql> CREATE TRIGGER trg_employee_insert
-> AFTER INSERT ON Employees
-> FOR EACH ROW
-> BEGIN
->     INSERT INTO Employee_Audit
->     (EmpID, Fname, Lname, Dname, DOJ, Salary, ActionType)
->     VALUES
->     (NEW.EmpID, NEW.Fname, NEW.Lname, NEW.Dname,
->     NEW.Doj, NEW.Salary, 'INSERT');
-> END$$
mysql> DELIMITER ;
```

Output:

```
Query OK, 0 rows affected (0.01 sec)
```

3b. AFTER UPDATE Trigger

Command:

```
mysql> DELIMITER $$
mysql> CREATE TRIGGER trg_employee_update
-> AFTER UPDATE ON Employees
-> FOR EACH ROW
-> BEGIN
->     INSERT INTO Employee_Audit
->     (EmpID, Fname, Lname, Dname, DOJ, Salary, ActionType)
->     VALUES
->     (NEW.EmpID, NEW.Fname, NEW.Lname, NEW.Dname,
->     NEW.Doj, NEW.Salary, 'UPDATE');
-> END$$
mysql> DELIMITER ;
```

Output:

```
Query OK, 0 rows affected (0.00 sec)
```

3c. AFTER DELETE Trigger

Command:

```
mysql> DELIMITER $$
mysql> CREATE TRIGGER trg_employee_delete
-> AFTER DELETE ON Employees
-> FOR EACH ROW
-> BEGIN
->     INSERT INTO Employee_Audit
->     (EmpID, Fname, Lname, Dname, DOJ, Salary, ActionType)
->     VALUES
->     (OLD.EmpID, OLD.Fname, OLD.Lname, OLD.Dname,
```

```

-> OLD.Doj, OLD.Salary, 'DELETE');
-> END$$
mysql> DELIMITER ;

```

Output:

```
Query OK, 0 rows affected (0.01 sec)
```

Step 4: Insert Employee Records

Command:

```

mysql> INSERT INTO Employees (Fname, Lname, Dname, DOJ, Salary) VALUES
-> ('Rama', 'Shastri', 'IT', '2026-05-23', 200000),
-> ('Amit', 'Sharma', 'HR', '2023-03-15', 50000),
-> ('Sneha', 'Patil', 'IT', '2022-07-10', 75000);

```

Output:

```

Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0

```

Step 5: Verify Data After INSERT

5a. View Employees Table

Command:

```
mysql> SELECT * FROM Employees;
```

Output:

EmpID	Fname	Lname	Dname	DOJ	Salary
1	Rama	Shastri	IT	2026-05-23	200000.00
2	Amit	Sharma	HR	2023-03-15	50000.00
3	Sneha	Patil	IT	2022-07-10	75000.00

3 rows in set (0.00 sec)

5b. View Employee_Audit Table (After INSERT)

Command:

```
mysql> SELECT * FROM Employee_Audit;
```

Output:

AuditID	EmpID	Fname	Lname	Dname	DOJ	Salary	ActionType	ActionDate
1	1	Rama	Shastri	IT	2026-05-23	200000.00	INSERT	2026-04-20 16:46:38

```
|      2 |      2 | Amit | Sharma | HR      | 2023-03-15 | 50000.00 | INSERT |
2026-04-20 16:46:38 |
|      3 |      3 | Sneha | Patil   | IT      | 2022-07-10 | 75000.00 | INSERT |
2026-04-20 16:46:38 |
+-----+-----+-----+-----+-----+-----+
+-----+
3 rows in set (0.00 sec)
```

Step 6: Create Stored Procedure with Cursor (Increment_IT_Salaries)

Command:

```
mysql> DELIMITER $$
mysql> CREATE PROCEDURE Increment_IT_Salaries()
-> BEGIN
->     DECLARE done          INT DEFAULT 0;
->     DECLARE v_EmpID       INT;
->     DECLARE v_Dname       VARCHAR(20);
->     DECLARE v_Salary      DECIMAL(10,2);
->
->     DECLARE emp_cursor CURSOR FOR
->         SELECT EmpID, Dname, Salary FROM Employees;
->
->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
->
->     OPEN emp_cursor;
->
->     read_loop: LOOP
->         FETCH emp_cursor INTO v_EmpID, v_Dname, v_Salary;
->
->         IF done THEN
->             LEAVE read_loop;
->         END IF;
->
->         IF v_Dname = 'IT' THEN
->             UPDATE Employees
->             SET Salary = v_Salary * 1.10
->             WHERE EmpID = v_EmpID;
->         END IF;
->
->     END LOOP;
->
->     CLOSE emp_cursor;
-> END$$
mysql> DELIMITER ;
```

Output:

```
Query OK, 0 rows affected (0.00 sec)
```

Step 7: Execute the Stored Procedure

Command:

```
mysql> CALL Increment_IT_Salaries();
```

Output:

```
Query OK, 0 rows affected (0.01 sec)
```

Step 8: Verify Data After Cursor Execution

8a. View Updated Employees Table

Command:

```
mysql> SELECT * FROM Employees;
```

Output:

EmpID	Fname	Lname	Dname	DOJ	Salary
1	Rama	Shastri	IT	2026-05-23	220000.00
2	Amit	Sharma	HR	2023-03-15	50000.00
3	Sneha	Patil	IT	2022-07-10	82500.00

3 rows in set (0.00 sec)

8b. View Employee_Audit Table (After UPDATE)

Command:

```
mysql> SELECT * FROM Employee_Audit;
```

Output:

AuditID	EmpID	Fname	Lname	Dname	DOJ	Salary	ActionType
1	1	Rama	Shastri	IT	2026-05-23	200000.00	INSERT
2	2	Amit	Sharma	HR	2023-03-15	50000.00	INSERT
3	3	Sneha	Patil	IT	2022-07-10	75000.00	INSERT
4	1	Rama	Shastri	IT	2026-05-23	220000.00	UPDATE
5	3	Sneha	Patil	IT	2022-07-10	82500.00	UPDATE

5 rows in set (0.00 sec)